INCH-POUND

MIL-PRF-1/10J
14 April 2005
SUPERSEDING
MIL-E-1/10H
17 February 1981

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5751

This specification sheet is inactive for new design after 11 August 2000.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Twin triode, miniature, high Mu

Outline - - - 6-2 (EIA)
Base - - - E9-1
Envelope - - - T6-1/2
Cathode - - - Coated unipotential
Base connections:

| Pin No. | - - - | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| Element | - - - | 2a | 2g | 2k | h | h | 1a | 1g | 1k | hct |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Eb | Ec | Ehk | Rg/g | Ik/k | Pp/p | TE | Alt |
|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | v | MegΩ | mAdc | W | °C | ft |
| Maximum: | 13.8 | 330 | 0, -50 | 100 | 0.5 | 6 | 0.8 | +165 | (see note 1) |
| Maximum: | 6.9 | --- | --- | --- | --- | --- | --- | --- | --- |
| Minimum: | 11.4 | --- | --- | --- | --- | --- | --- | --- | --- |
| Minimum: | 5.7 | --- | --- | --- | --- | --- | --- | --- | --- |
| Test conditions: | 12.6 | 250 | -3 | 0 | --- | --- | --- | --- | |

GENERAL:

Qualification: Not required.

Reliable tube

AMSC N/A FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

TABLE I. Test conditions.

| MIL-STD-1311 | Requirement or test | Conditions | Acceptance level (see note 11) | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Method | Conformance inspection, part 1 | | | | | | |
| 1301 | Heater current | | 0.4 | If | 160 | 190 | mA |
| 1336 | Heater-cathode leakage | See note 2 | 0.4 | Ihk | --- | 10 | µAdc |
| 1266 | Total grid current | Rg/g = 0.5 MegΩ (see notes 2 and 3) | 0.4 | Ic | 0 | -0.4 | µAdc |
| 1256 | Electrode current (1) (anode) | See note 2 | 0.4 | Ib | 0.4 | 1.8 | mAdc |
| 1256 | Electrode current (2) (anode) | Ec = -10.5 Vdc Rp = 0.1 MegΩ (see note 2) | 0.4 | Ib | --- | 10 | µAdc |
| 1321 | AC amplification | Ebb = 100 Vdc; Ec = 0; Esig = 0.2 V ac; Rp = 0.5 MegΩ; Rg = 10 MegΩ (see note 2) | 0.4 | Ep | 7.5 | --- | Vac |
| 1201 | Short and discontinuity detection | | 0.4 | --- | --- | --- | --- |
| | Conformance inspection, part 2 | | | | | | |
| 1211 | Insulation of electrodes | See note 2 | 2.5 | R | 500 | --- | MegΩ |
| 1256 | Electrode current (1) (anode) (difference between sections) | | 2.5 | Ib | --- | 0.6 | mAdc |
| 1306 | Transconductance (1) | See note 2 | 2.5 | Sm | 900 | 1600 | µmhos |
| 1306 | Transconductance (2) | Ef = 11.4 V (see note 2) | 2.5 | $\frac{\Delta Sm}{Ef}$ | --- | 15 | % |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

TABLE I. Test conditions - Continued.

| MIL-STD-1311 | Requirement or test | Conditions | Acceptance level (see note 11) | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Method | Conformance inspection, part 2 -Continued | | | | | | |
| 1266 | Grid emission | Ef = 15.0 V; R/g = 0.5 MegΩ; Ec = -12.0 Vdc (see notes 2 and 4) | 2.5 | Ic | 0 | -0.6 | µAdc |
| 2201 | Noise and microphonics | Ef = 12.6 Vac; Ebb = 300 Vdc; Ec = 0; Ecal = 60 mVac; Rk = 1,500 ohms; Rp = 0.1 MegΩ (see note 5) | 2.5 | --- | --- | --- | --- |
| 1316 | Amplification factor | See note 2 | 6.5 | Mu | 55 | 85 | --- |
| 1331 | Direct-interelectrode capacitance | No shield (note 2) | 6.5 | Cgp | 1.10 | 1.70 | pF |
| | | No shield (note 2) | | Cin | 1.10 | 1.70 | pF |
| | | No shield, sec 1 | | Cout | 0.23 | 0.69 | pF |
| | | No shield, sec 2 | | Cout | 0.19 | 0.53 | pF |
| 1031 | High-frequency vibration | Rp = 2,000 ohms (see note 5) | 6.5 | Ep | --- | 100 | mVac |
| 1031 | Sweep-frequency vibration | Rp = 2,000 ohms (see note 5) | 6.5 | Ep | --- | 300 | mV ac |
| 1041 | Shock | 630 G; Ehk = +100 Vdc (see note 7) | 6.5 | --- | --- | --- | --- |
| 1031 | Vibration fatigue | | 6.5 | --- | --- | --- | --- |
| --- | Post-shock and vibration-fatigue test end points: | | | | | | |
| 1031 | Low-frequency vibration | | --- | Ep | --- | 150 | mVac |
| 1336 | Heater-cathode leakage | | --- | Ihk | --- | 20 | µAdc |
| 1321 | AC amplification | | --- | Ep | 6.5 | --- | Vac |
| 1266 | Total grid current | | --- | Ic | 0 | -0.6 | µAdc |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

TABLE I. Test conditions - Continued.

| MIL-STD-1311 | Requirement or test | Conditions | Acceptance level (see note 11) | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Method | Conformance inspection, part 2 -Continued | | | | | | |
| 1121 | Base strain | See note 8 | --- | --- | --- | --- | --- |
| 2126 | Envelope strain | | 2.5 | --- | --- | --- | --- |
| 1105 | Permanence of marking | | --- | --- | --- | --- | --- |
| MIL-STD-1311 | Requirement or test | Conditions | | Symbol | Limits | | Unit |
| | | | | | Min | Max | |
| | Conformance inspection, part 3 | | | | | | |
| 1506 | Heater-cycling life | Ef = 7.5 V; heaters in parallel; Ehk = +135 Vdc; Ec = Eb = 0 | | --- | --- | --- | --- |
| --- | Heater-cycling life-test end-point: | | | | | | |
| 1336 | Heater-cathode leakage | | | Ihk | --- | 15 | --- |
| 1516 | Stability life | Ehk = 135 V dc; Ec = -2 Vdc; Rg/g = 0.5 MegΩ; TA = room | | --- | --- | --- | --- |
| --- | Stability life-test end point (2 and 20 hours) | | | | | | |
| 1321 | Change in amplification of individual tubes | | | ΔEpt | --- | 10 | % |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

TABLE I. Test conditions - Continued.

| MIL-STD-1311 | Requirement or test | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| | Conformance inspection, part 3 - Continued | | | | | |
| 1501 | Intermittent life | Ehk = 135 Vdc; Ec = -2 Vdc; Rg/g = 0.5 MegΩ; TE = +165°C (min) (see notes 9 and 10) | --- | --- | --- | --- |
| --- | Intermittent life-test end-points (1,000 hours): | | | | | |
| --- | Inoperatives | | --- | --- | --- | --- |
| 1266 | Total grid current | | Ic | 0 | -0.4 | µAdc |
| 1301 | Heater current | | If | 160 | 196 | mA |
| 1321 | Change in AC amplification of individual tubes | | ΔEp | --- | 20 | % |
| 1336 | Heater-cathode leakage | | Ihk | --- | 10 | µAdc |
| 1211 | Insulation of electrodes | | R | 200 | --- | MegΩ |
| 1511 | Cathode interface life | Ef = 6.9 V; heaters in parallel (see note 2) | Ri | --- | 50 | Ohms |

NOTES:

1. See "Reduced pressure (altitude) rating" and altitude, maximum peak voltage.
2. Test each unit separately.
3. This test shall be performed at the conclusion of the holding period.
4. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions indicated below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

| Ef | Ec1 | Eb | Rk/k | Rg/g |
|---|---|---|---|---|
| V | Vdc | Vdc | Ohms | MegΩ |
| 15.0 | -3.0 | 250 | 0 | 0.5 |

5. Tie 1k to 2k; 1g to 2g; and 1a to 2a.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

TABLE I. Test conditions - Continued.

6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12-month rule shall apply.

7. A grid resistor of 0.1 megohm shall be added to each section; however, this resistor shall not be used when a thyratron-type short indicator is employed.

8. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (MIL-STD-1311, method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

   (a) Accepted if not more than one defective for class "A", "B", or "C" defects, respectively (see MIL-STD-1311, method 1121), or if not more than a total of two defectives are found in the sample.

   (b) Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.

9. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (MIL-STD-1311, method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

10. The life-test sample shall consist of 20 tubes per lot, and no failures shall be permitted. In the event of rejection due to failure of one or more tubes, a second sample of 20 tubes shall be selected from the lot. Acceptance shall then be based upon zero failures in the second sample.

11. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

NOTE: Referenced documents. In addition to MIL-PRF-1, this document references the following: MIL-STD-1311. Government documents are available via the ASSIST database at http://assist.daps.dla.mil.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR
Navy - AS, CG, MC
Air Force - 99

Preparing activity:
DLA - CC

(Project 5960-3757)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:46Z
Check the source to verify that this is the current version before use.